Mock Exam 3 — Version A

1. Curl, divergence, and conservativity

On all of $\mathbb{R}^3$, let

$$\mathbf{F}(x,y,z) = \big(y(\sin z - 1),\ x(\sin z + 1),\ xy\cos z\big).$$

Compute $\nabla \times \mathbf{F}$ and $\nabla \cdot \mathbf{F}$. Determine whether $\mathbf{F}$ is conservative, and justify your conclusion.

2. Green's theorem

Let D be the region

$$x^2 + y^2 \le 4, \qquad (x-1)^2 + y^2 \ge \frac{1}{4}.$$

Its boundary C consists of both circles, oriented so that D stays on your left as you traverse each component. Use **Green's theorem** to evaluate

$$\oint_C \big(ye^{xy} - y\big)\,\mathrm{d}x + \big(xe^{xy} + 3x^2 + x\big)\,\mathrm{d}y.$$

Show the double-integral setup and its bounds.

3. Three surface parametrizations

For each surface, give a parametrization using **exactly two parameters**, together with bounds for both parameters. No orientation is required.

(a) The portion of the cone $z = \sqrt{x^2 + y^2}$ inside the cylinder $(x-1)^2 + y^2 \le 1$.

(b) The portion of the ellipsoid

$$\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1$$

satisfying $y \ge 0$ and $z \ge \sqrt{\dfrac{x^2}{4} + \dfrac{y^2}{9}}$.

(c) The surface obtained by rotating the planar arc

$$(x,y) = (2 + \cos s, \sin s), \qquad -\frac{\pi}{2} \le s \le \frac{\pi}{2},$$

once about the y-axis.

4. A scalar surface integral: setup only

The surface S is parametrized by

$$\mathbf{r}(u,v) = (u+v, u-v, uv), \qquad u \ge 0, \quad v \ge 0, \quad u + v \le 1.$$

For

$$f(x,y,z) = e^{x^2 - y^2 - 4z} + z,$$

set up, but **do not evaluate**, the scalar surface integral $\displaystyle\iint_S f\,\mathrm{d}\sigma$. Show the two tangent vectors, their cross product, its magnitude, the substituted scalar function, and the final double integral with bounds.

5. Direct flux through a curved surface

Let S be the portion of the paraboloid

$$z = x^2 + y^2, \qquad 1 \le z \le 4,$$

oriented **outward, away from the z-axis**. Let

$$\mathbf{F}(x, y, z) = \big(x - ye^{z^2},\ y + xe^{z^2},\ 2z - e^z\big).$$

Set up and evaluate $\displaystyle\iint_S \mathbf{F} \cdot \mathbf{N}\, d\sigma$ by parametrizing S directly. Show your tangent cross product, check its orientation, substitute the field, and evaluate the resulting double integral. Include only the curved surface; do not add end caps.

6. Stokes' theorem and the whole boundary

Let S be the portion of the cone

$$z = \sqrt{x^2 + y^2}, \qquad 1 \le z \le 3,$$

oriented **outward, away from the z-axis**, and let

$$\mathbf{F}(x, y, z) = \big(-(z+1)y,\ (z+1)x,\ e^{z(x^2+y^2)}\big).$$

Use **Stokes' theorem** to evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, d\sigma.$$

Identify every boundary component and its induced direction. Use one parameter on each component, show the boundary line-integral setup, and evaluate it.

7. Conceptual decisions

(a) Let f be a scalar field and $\mathbf{G}$ a vector field, both with continuous second partial derivatives on $\mathbb{R}^3$. For each expression, state whether it is a scalar field, a vector field, or undefined. For each defined expression, also state whether it is *necessarily* zero.

$$\text{(i) } \nabla \times (f\mathbf{G}), \qquad \text{(ii) } \nabla \cdot (\nabla \times \mathbf{G}), \qquad \text{(iii) } \nabla \times (\nabla \cdot \mathbf{G}).$$

(b) Let $\mathbf{F} = (0, xy, 0)$ and let C be $\mathbf{r}(t) = (t, 0, 0)$, $0 \le t \le 1$. Select all methods that apply *directly to* $\int_C \mathbf{F} \cdot d\mathbf{r}$ *with the stated field and curve*, and give a one-sentence justification:

direct parametrization, FTLI, Green's theorem, Stokes' theorem.

Here FTLI means the Fundamental Theorem of Line Integrals. Do not add auxiliary curves or surfaces. In particular, consider whether knowing that the curl vanishes *on* C is sufficient for FTLI.

(c) At a point p in a fluid velocity field,

$$(\nabla \cdot \mathbf{V})(p) = -2, \qquad (\nabla \times \mathbf{V})(p) = (0, 0, 4).$$

Does the divergence indicate a local source or sink? Give the directed axis of local rotation and its clockwise/counterclockwise sense when viewed from the positive z-axis.

Mock Exam 3 — Version B

1. Curl, divergence, and conservativity

Let the vector field below be defined on all of $\mathbb{R}^3$:

$$\mathbf{F}(x,y,z) = \big(ye^{xy} + z\cos x - yz,\ xe^{xy} + z^2 + xz,\ \sin x + 2yz\big).$$

Calculate $\nabla \times \mathbf{F}$ and $\nabla \cdot \mathbf{F}$. Determine whether $\mathbf{F}$ is conservative, and justify your conclusion.

2. Green's theorem on an unfinished curve

Let

$$\mathbf{F}(x,y) = \left(ye^{xy} - \frac{y^2}{2},\ xe^{xy} + 2x\right).$$

The path C follows these pieces in order:

- the straight segment from $(1,0)$ to $(0,0)$;
- the parabola $x = y^2$ from $(0,0)$ to $(1,1)$;
- the straight segment from $(1,1)$ to $(2,1)$.

Use **Green's theorem** to evaluate

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}.$$

Show the closed curve and region to which you apply the theorem, and account for any additional path you introduce.

3. Surface parametrizations and intersection bounds

For each surface, give a parametrization using **exactly two parameters**, together with bounds for both parameters. Include only the specified surface, without any additional caps.

(a) The part of the sphere $x^2 + y^2 + z^2 = 4$ that is outside the cylinder $x^2 + y^2 = 1$ and satisfies $z \geq 0$.

(b) The part of the cylinder $x^2 + y^2 = 1$ between the planes $z = y$ and $z = 2 + x$.

(c) The part of the cone $z = \sqrt{x^2 + y^2}$ inside the sphere $x^2 + y^2 + z^2 = 4$ and at or above the plane $z = 1$.

4. Set up a scalar surface integral

A surface S is parametrized by

$$\mathbf{r}(u,v) = (u, u^2 + v, uv), \qquad \begin{cases} 1 \leq u \leq 2, \\ 0 \leq v \leq 2 - u. \end{cases}$$

Set up, but **do not evaluate**, the scalar surface integral

$$\iint_S f(x,y,z)\ \mathrm{d}\sigma, \qquad f(x,y,z) = e^{\,y - x^2} + \left(\frac{z}{x}\right)^2.$$

Show the tangent vectors, their cross product, its magnitude, the substitution into f, and the complete double integral.

5. Flux through a curved sheet

Let S be the portion of the graph $z = xy$ above the planar region enclosed by $y = x^2$ and $x = y^2$ in the first quadrant. Orient S **downward**. For

$$\mathbf{F}(x, y, z) = (x + 1, y - 1, 3z + x - y),$$

set up and evaluate

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, d\sigma.$$

Use a direct surface parametrization. Show the cross product, check its orientation, substitute the field, and give the complete double integral.

6. Stokes' theorem with a nonplanar boundary

Let S be the graph

$$z = x^2 - y^2, \qquad x^2 + y^2 \le 1,$$

oriented upward. Define

$$\begin{aligned}\mathbf{F}(x, y, z) = \big(& -yz^2 + (1 - x^2 - y^2)e^{xyz}, \\ & xz^2 + (1 - x^2 - y^2)\sin(e^{xz}), \\ & e^{x^2+y^2} + ze^z\big).\end{aligned}$$

Use **Stokes' theorem** to evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma.$$

Identify and orient the boundary, give a one-parameter parametrization, and evaluate the resulting line integral.

7. Operators, physical meaning, and choice of method

Assume that all scalar and vector fields in this problem are smooth on $\mathbb{R}^3$.

(a) For each expression, state whether it is a scalar field, a vector field, or undefined. Here f is scalar and $\mathbf{F}$ is vector-valued.

$$\begin{aligned}&\text{(i) } \nabla \times \big((\nabla f) \times \mathbf{F}\big), \quad \text{(ii) } (\nabla f) \cdot (\nabla \times \mathbf{F}), \\ &\text{(iii) } \nabla \times (\nabla \cdot \mathbf{F}), \qquad \text{(iv) } \nabla \cdot (\nabla \times \mathbf{F}).\end{aligned}$$

(b) For $\mathbf{V} = (-y, x, z)$ near the origin, determine whether a small fluid element experiences local volume expansion, compression, or neither. Determine whether its local rotation is clockwise, counterclockwise, or neither as viewed from the positive z-axis. Justify both answers with divergence and curl.

(c) Consider $\mathbf{W} = (2xy, x^2, z)$ along

$$\mathbf{c}(t) = (\cos t, \sin t, 0), \qquad 0 \le t \le 2\pi.$$

Which of the following methods can be applied **directly, without adding any auxiliary path**, to evaluate $\int_C \mathbf{W} \cdot d\mathbf{r}$? Select all that apply and briefly justify the selections and exclusions: direct parametrization; Fundamental Theorem of Line Integrals; Green's theorem; Stokes' theorem. For Green's theorem, use the planar field formed by the first two components on $z = 0$.

Mock Exam 3 — Version C

1. Curl, divergence, and conservativity

Let $D = 1 + x^2 + y^2 + z^2$ and define the field on all of $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) = \left(yz + \frac{2x}{D},\ xz + \frac{2y}{D},\ xy + \frac{2z}{D} \right).$$

Calculate $\nabla \times \mathbf{F}$ and $\nabla \cdot \mathbf{F}$. Determine whether $\mathbf{F}$ is conservative and justify your conclusion.

2. Green's theorem

Let C be the positively oriented boundary of

$$R = \{(x, y) : x^2 + y^2 \le 2x,\ y \ge 0\}.$$

Use **Green's theorem** to evaluate

$$\oint_C ye^{xy}\ \mathrm{d}x + \left(xe^{xy} + \frac{x^3}{3} + xy^2 \right) \mathrm{d}y.$$

Show the scalar curl, the region and bounds for your double integral, and its evaluation.

3. Surface parametrizations

Give a parametrization with **exactly two parameters**, including the bounds for both parameters, for each surface.

(a) The portion of the ellipsoid

$$\frac{x^2}{9} + \frac{y^2}{4} + z^2 = 1$$

that satisfies $z \ge 0$ and $x^2/9 + y^2/4 \le 1/4$.

(b) The portion of the cone $z = \sqrt{x^2 + y^2}$ in the first octant that lies above $z = y + 1$ and below $z = 2$.

(c) The ruled surface formed by joining the corresponding points

$$\mathbf{a}(s) = (s, 0, s), \qquad \mathbf{b}(s) = (0, s, 2), \qquad 0 \le s \le 1,$$

by straight line segments.

4. Scalar surface integral: setup only

The surface S is given by

$$\mathbf{r}(u, v) = (u\cos v,\ u\sin v,\ uv), \qquad \begin{cases} 1 \le u \le 2, \\ 0 \le v \le u - 1. \end{cases}$$

For

$$f(x, y, z) = (x^2 + y^2) \exp\left(\frac{z}{\sqrt{x^2 + y^2}} \right),$$

set up, but **do not evaluate**, $\displaystyle\iint_S f\ \mathrm{d}\sigma$. Show both tangent vectors, their cross product and its magnitude, and the substituted scalar function before writing your integral.

5. Direct computation of oriented flux

Let S be the curved cylindrical wall

$$(x-z)^2+y^2=1, \qquad 0\le z\le 2+y.$$

Orient S outward, away from the axis $x=z$, $y=0$, and include **only the curved wall**. Let

$$\mathbf{F}(x,y,z)=\left(z(x-z)+e^{x+y+z},\ yz,\ e^{x+y+z}\right).$$

By **direct surface parametrization**, set up and evaluate

$$\iint_S \mathbf{F}\cdot\mathbf{N}\,\mathrm{d}\sigma.$$

Show a parametrization, the oriented cross product, a check of its direction, the substituted field, the dot product, and the integration bounds.

6. Stokes' theorem

Let S be the graph $z=xy$ over the triangle

$$D=\{(x,y): x\ge 0,\ y\ge 0,\ x+y\le 1\},$$

oriented **downward**. For

$$\mathbf{F}(x,y,z)=\left(ye^{xy}+yz^2,\ xe^{xy}+xz^2+z,\ 2xyz\right),$$

use **Stokes' theorem** to evaluate

$$\iint_S (\nabla\times\mathbf{F})\cdot\mathbf{N}\,\mathrm{d}\sigma.$$

Identify the induced boundary and give a one-parameter parametrization of it, which may be piecewise. Show the derivative, substitution, dot product, and evaluation of the boundary line integral.

7. Conceptual questions

(a) Let f be a C^3 scalar function and $\mathbf{V}$ a C^2 vector field on $\mathbb{R}^3$. For each expression, state whether it is a scalar, a vector, or undefined. For a defined expression, also state whether it must always be zero.

$$\text{(i) } \nabla\cdot\big(\nabla\times(f\mathbf{V})\big), \qquad \text{(ii) } \nabla\times(\nabla\cdot\mathbf{V}), \qquad \text{(iii) } \nabla\cdot(f\nabla f).$$

(b) For $\mathbf{H}=(yz,xz,xy)$ and the curve

$$\mathbf{c}(t)=(t,t^2,t^3), \qquad 0\le t\le 1,$$

select **all** methods that apply directly to $\displaystyle\int_C \mathbf{H}\cdot\mathrm{d}\mathbf{r}$ as stated, without adding curves or changing the field. Briefly justify your choices.

A. Direct parametrization
B. Fundamental Theorem of Line Integrals
C. Green's theorem
D. Stokes' theorem

(c) A velocity field has, at a point P,

$$\mathbf{v}(P)=\mathbf{0}, \qquad (\nabla\times\mathbf{v})(P)=(2,0,0), \qquad (\nabla\cdot\mathbf{v})(P)=0.$$

Give the axis and sense of its local rotational tendency, viewed from the positive x-axis, and state whether its divergence indicates a local source, a local sink, or neither at P.

Challenge Mock Exam 3 — Version D

Intentionally harder than the actual Exam 3. Work untimed and justify your answers.

1. Curl, divergence, and global conservativity

For a real constant λ, let

$$q = (x-z)^2 + 2y^2, \qquad U = \{(x,y,z) : q > 0\},$$

$$\mathbf{F}_\lambda(x,y,z) = \left(y - \frac{\lambda y}{q}, \quad x + \frac{\lambda(x-z)}{q}, \quad 2z + \frac{\lambda y}{q} \right).$$

Calculate $\nabla \times \mathbf{F}_\lambda$ and $\nabla \cdot \mathbf{F}_\lambda$ on U. Find all λ for which the field is conservative on *all of* U. Give a potential when one exists and a decisive justification when one does not.

2. Green's theorem

Let C be the positively oriented boundary of

$$\begin{cases} 0 \le x \le 1, \\ 0 \le y \le \ln(1+x). \end{cases}$$

Use **Green's theorem** to evaluate exactly

$$\oint_C \left(y e^{xy} - \frac{y}{1+x^2} \right) \mathrm{d}x + x e^{xy} \, \mathrm{d}y.$$

Show the scalar curl, the double integral with bounds, and its evaluation.

3. Three surface parametrizations

Give a parametrization with **exactly two parameters**, and complete bounds for both, for each surface. No orientation is required.

(a) The surface obtained by rotating the arc

$$\mathbf{c}(t) = (t, 0, t^2), \qquad 1 \le t \le 2,$$

once about the line $L = \{(a,a,0) : a \in \mathbb{R}\}$.

(b) The portion of $x^2 + y^2 - z^2 = 1$ with $0 \le z \le 1$. Your parametrization must have the additional property that holding its first parameter fixed traces a **straight line segment** as the second parameter varies.

(c) The portion of the sphere $x^2 + y^2 + z^2 = 4$ satisfying $z \ge |x| + |y|$.

4. A scalar surface integral: setup only

Let S be the image of

$$\mathbf{r}(u,v) = (u^2 - v^2, 2uv, u^2 + v^2), \qquad 1 \le u^2 + v^2 \le 4.$$

For $f(x,y,z) = \sqrt{z+x} + e^z$, set up, but **do not evaluate**, $\displaystyle\iint_S f \, \mathrm{d}\sigma$, counting each point of the *geometric surface* once. Show the tangent vectors, their cross product and its magnitude, the substituted function, and explicit integration bounds. Account for any repeated coverage by the supplied map.

5. Direct flux

Let S be the portion of

$$z = 1 + x^2 + y^2, \qquad \frac{1}{4} \le x^2 + y^2 \le 4,$$

oriented **upward**. For

$$\mathbf{H}(x, y, z) = \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}},$$

set up and evaluate $\iint_S \mathbf{H} \cdot \mathbf{N} \, d\sigma$ by **direct surface parametrization**. Show the oriented cross product, check its direction, substitute the field, and evaluate the resulting double integral. Include only the stated curved surface.

6. Stokes' theorem

Let

$$S = \{(x, y, z) : x^2 + y^2 + z^2 = 4, \ z \ge 0, \ (x - 1)^2 + y^2 \le 1\},$$

oriented outward from the sphere, and let $\mathbf{F} = (0, z^2, -yz)$. Use **Stokes' theorem** to evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma.$$

Identify the entire boundary and its induced direction. Give a one-parameter boundary parametrization, its derivative, the substituted field, the dot product, and the evaluated line integral.

7. Operators, physical meaning, and a global obstruction

(a) Let f be a C^2 scalar field and $\mathbf{G}$ a C^2 vector field on $\mathbb{R}^3$. Is $\nabla \cdot (f \nabla \times \mathbf{G})$ a scalar field, a vector field, or undefined? Must it always be zero? Give a general formula and either a justification of zero or an explicit counterexample.

(b) Regard the field $\mathbf{H}$ from Problem 5 as a velocity field on $U_0 = \mathbb{R}^3 \setminus \{(0, 0, 0)\}$. Calculate its divergence and curl. At a point of U_0, do they indicate local volume expansion, compression, or neither, and a local rotational tendency or none? Reconcile your answer with the fact that $\mathbf{H}$ points outward on every sphere centered at the origin.

(c) Does there exist a C^2 vector field $\mathbf{A}$ on *all of* U_0 such that $\nabla \times \mathbf{A} = \mathbf{H}$? Justify your answer using **Stokes' theorem** and a suitable surface integral. No divergence theorem is needed.